A2 Warm Up Compound Interest & Graphing Exp and Log Equations

Remember not to round until the very end

1. Bill invests $2977 in a retirement account with a fixed annual interest rate of 6% compounded quarterly. What will the account balance be after 13 years, rounded to the nearest hundredths.

P $\quad$ $6\% = .06$ $\quad$ $n = 4$ $\quad$ $t = 13$

$$A = 2977\left(1+\left(\frac{.06}{4}\right)\right) \boxed{x^{\square}} \ (4 \times 13) \boxed{=}$$

$$= \boxed{\$6{,}456.74}$$

2. Talia invests $2476 in a savings account with a fixed annual interest rate of 4% compounded monthly. What will the account balance be after 6 years, rounded to the nearest cent?

$4\% = .04$ $\quad$ $n = 12$

$$A = 2476\left(1+\left(\frac{.04}{12}\right)\right)^{(12 \times 6)}$$

$$= \boxed{\$3{,}146.36}$$

3. Graph $y = 3(2)^x - 4$ *remember*: asymptote, 2 special points (exp. = 0, exp = 1), and shape

b>1 and exponential growth!

BASIC SHAPE

asymptote @ $y = -4$

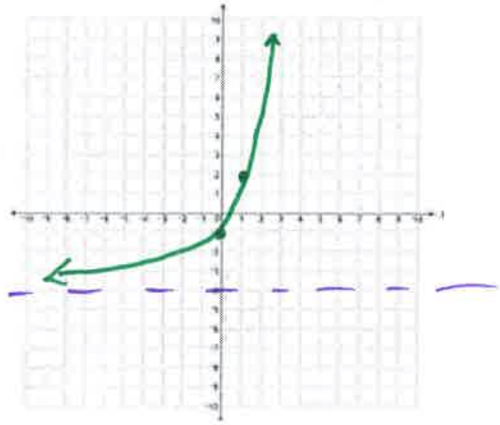
exp = 0 $\quad$ exp = 1

$x = 0$ $\quad$ $x = 1$

$y = 3(2)^0 - 4$ $\quad$ $y = 3(2)^1 - 4$

$y = 3 - 4 = -1$ $\quad$ $= 2$

$(0,-1)$ $\quad$ $(1,2)$



4. Graph $y = \log_{10}(x + 3) - 2$ asymptote, 2 special points: ( ) = base, ( ) = 1, and shape

log graph! Basic shape:

Asymptote $x = -3$

( ) = base $\quad$ ( ) = 1

$x + 3 = 10$ $\quad$ $x + 3 = 1$

$x = 7$ $\quad$ $x = -2$

$y = \underbrace{\log_{10}(7+3)}$ -2 $\quad$ $y = \log_{10}(-2+3) - 2$

$y = 1 - 2$ $\quad$ $= \underbrace{\log_{10}(1)} - 2$

$y = -1$ $\quad$ $= 0 - 2 = -2$

$(7,-1)$ $\quad$ $(-2,-2)$